Password Manager — Feature Plan

Beginner-Friendly Cybersecurity Hackathon Project

Project Overview

A web-based password manager that lets registered users securely store and manage their passwords. Each user has their own encrypted vault. The app includes four features: a real-time strength checker, a duplicate detector, a password expiry tracker, and a breach check using the free Have I Been Pwned API.

Database Structure

Users Table: user_id, username, email, hashed_password (bcrypt), created_at

Password Entries Table: entry_id, user_id (FK), site_name, site_url, username_for_site, encrypted_password (AES), expiry_date, breach_status, created_at, updated_at

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Feature 1: Password Strength Checker

Analyzes passwords in real time and gives a score based on length, complexity, and common patterns.

CRUD Operations

- **Create** — Submit a password for analysis; system generates a strength score and feedback.
- **Read** — View the score breakdown: length rating, character variety, common pattern warnings, and entropy estimate.

How It Works

Check length (8+ good, 12+ strong, 16+ excellent). Check character types (uppercase, lowercase, digits, symbols). Flag common patterns (dictionary words, sequences like '123', keyboard patterns like 'qwerty'). Calculate entropy and output a 0–100 score with color indicator.

Security note: All checking happens client-side. The plaintext password is never sent to the server for this feature.

Feature 2: Duplicate Password Detector

Scans all entries in the user's vault and flags any passwords that are reused across different sites.

CRUD Operations

- **Read** — Scan and compare all saved entries; display grouped duplicates with a suggestion to change them.

How It Works

Decrypt all entries for the logged-in user. Compare hashes (not plaintext) to find matches. Group duplicates together and display which sites share the same password. Show a warning severity level based on how many sites overlap.

Why only Read: This is a scan-and-report feature. The user decides whether to act. No data is created or modified by the scan itself.

Feature 3: Password Expiry Tracker

Users set an expiry period for each saved password. The system flags passwords that are old and due for a change.

CRUD Operations

- **Create** — Set an expiry duration when saving a password (e.g., 30, 60, or 90 days).
- **Read** — View a dashboard showing which passwords are expired, expiring soon, or still fresh.
- **Update** — Reset the expiry timer when the user changes their password.

How It Works

Store created_at and expiry_days for each entry. Calculate days remaining = expiry_days - (today - created_at). Categorize: Expired (0 or less), Expiring Soon (under 7 days), Fresh (more than 7 days). Display as a color-coded dashboard (red / yellow / green).

Why it matters: Stale passwords are a real security risk. This teaches date-based logic and promotes good security hygiene.

Feature 4: Breach Check per Entry

Checks whether each saved password has appeared in known data breaches using the free Have I Been Pwned Passwords API.

CRUD Operations

- **Create** — Automatically run a breach check when a new password is added to the vault.
- **Read** — View breach status for all entries: Safe, Compromised, or Unchecked.
- **Delete** — Clear stored breach results when a user updates their password and wants a fresh check.

How It Works

Hash the password with SHA-1. Send only the first 5 characters of the hash to the HIBP API (k-anonymity). API returns all matching suffixes. Check locally if the full hash is in the returned list. Store result as breach_status on the entry.

The Pwned Passwords API is completely free — no API key or subscription needed. The full password never leaves the user's device.

CRUD Summary Table

Feature	C	R	U	D
1. Strength Checker	✓ Check a password	✓ View score & feedback		

2. Duplicate Detector		✓ Scan & compare entries		
3. Expiry Tracker	✓ Set expiry duration	✓ View status dashboard	✓ Reset timer	
4. Breach Check	✓ Run check on add	✓ View breach status		✓ Clear old results